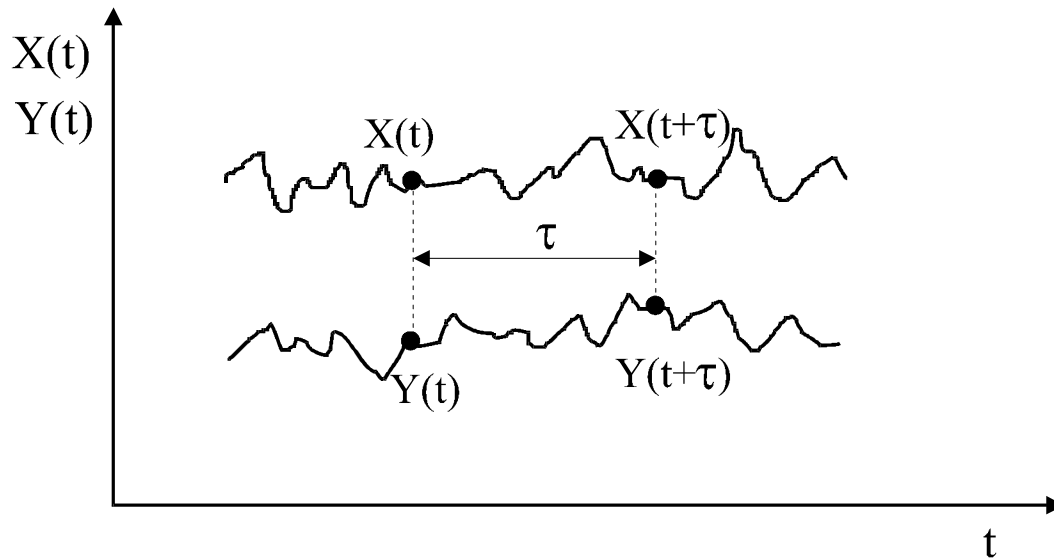


Exercise 4

Application of cross-correlation function to identify signal properties



cross-correlation functions

$$R_{xy}(\tau) = E[X(t) \cdot Y(t + \tau)] = \overline{X(t) \cdot Y(t + \tau)}$$

$$R_{yx}(\tau) = E[Y(t) \cdot X(t + \tau)] = \overline{Y(t) \cdot X(t + \tau)}$$

cross-covariance functions

$$C_{xy}(\tau) = E[x(t) \cdot y(t + \tau)] = \overline{x(t) \cdot y(t + \tau)}$$

$$C_{yx}(\tau) = E[y(t) \cdot x(t + \tau)] = \overline{y(t) \cdot x(t + \tau)}$$

$$x(t) = X(t) - \bar{X}$$

$$y(t) = Y(t) - \bar{Y}$$

normalized cross-covariance function

$$Q_{xy}(\tau) = \frac{C_{xy}(\tau)}{\sigma_X \cdot \sigma_Y} \in \langle -1; 1 \rangle$$

cross-covariance function estimator

$$C_{xy}(\tau) = C_{xy}(i \cdot \Delta t) = \frac{1}{N-i} \sum_{j=1}^{j=N-i} x_j \cdot y_{j+i}$$

Compulsory tasks

- a) perform stationarity test of cross-covariance function for compound signals with strong noise content (at least 50%) having the same period of deterministic components; show the evolution of the correlation function for the varying signal length
- b) analyse the relationships between amplitudes of correlated signals and correlation function for various deterministic signals of the same periods; repeat the analysis for "disturbed" signals, i.e. with added noise; refer to the lecture on correlation analysis, slide No 16, 17



please distinguish the difference (diagnostic value) between dimensional and normalised correlation functions

- c) analyse the influence of signals' phase angles on the correlation function distribution for deterministic (or compound) signals of the same periods; refer to the lecture on correlation analysis, slide No 16, 17
- d) apply the cross-correlation function to the analysis of the "beat" phenomenon, i.e. combination of 2 sine waves of slightly different frequencies

$$f_1, f_2 = \bar{f} \pm \frac{\Delta f}{2}; \quad \Delta f \ll \bar{f}$$

$$\bar{f} = \frac{f_1 + f_2}{2}$$

provide the recipe on how the original frequencies f_1, f_2 can be found from the correlogram; refer to the lecture on correlation analysis, slide No 19

Optional tasks

- analyse the relationship between the correlation functions $Q_{xy}(\tau)$ and $Q_{yx}(\tau)$ for the selected pair of periodic signals
- choose the maximum correlation distance for 2 signals of significantly different time scales (periods)
-

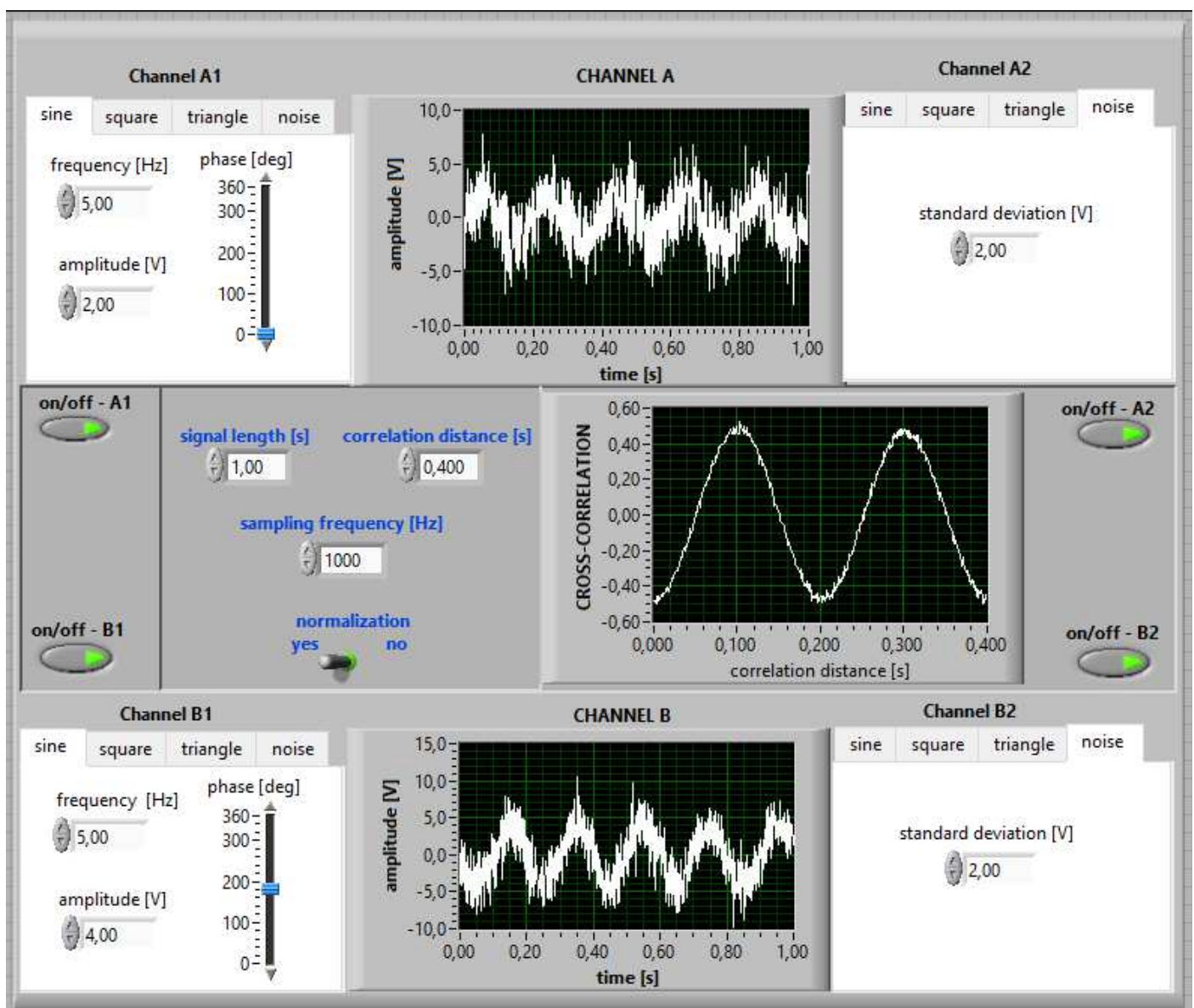


The exercise aims at learning how to:

- recognise the correlated signal types,
- find their parameters

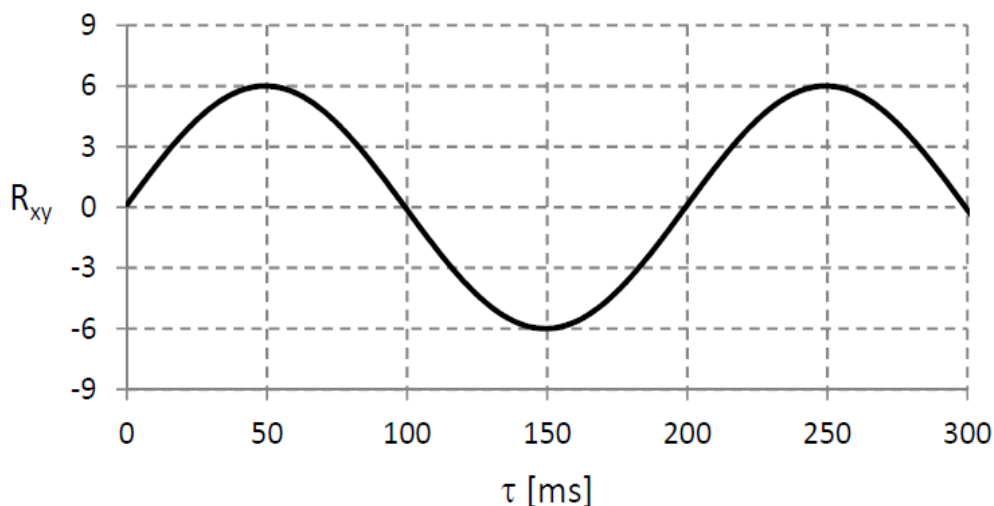
having known cross-correlation function distribution (and, of course, not seeing the signals).

project "correlation" in LabView environment



Sample problems

1. What should be the relationship between max. correlation distance and signal duration ensuring the accurate estimation of correlation function?
2. What is the unit of the cross-correlation function and what is the range of values it may take ?
3. What is the unit of the normalized cross-covariance function and what is the range of values it may take?
4. Can the correlation function be periodic ? If so, when ?
5. Describe (by drawing or in written form) the distribution of the cross-covariance function of two sine waves of the same frequency, shifted by phase angle 90° . Indicate the relationships between signals parameters and correlation function.
6. Where (at which correlation distance) is the cross-correlation function maximum located ? What level does the maximum take ?
7. Having known the distribution of cross-correlation function describe the signals, i.e. provide their compositions and parameters (if possible).



8. Having known the distribution of cross-correlation function describe the signals, i.e. provide their compositions and parameters (if possible).

